



# CrossLinkU-NX Evaluation Board User Guide

## User Guide

FPGA-EB-02062-1.0

October 2023

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## Acronyms in This Document

A list of acronyms used in this document.

| Acronym          | Definition                              |
|------------------|-----------------------------------------|
| CMOS             | Complementary Metal-Oxide Semiconductor |
| CSI-2            | Camera Serial Interface                 |
| DDR              | Double Data Rate                        |
| DSI              | Display Serial Interface                |
| FTDI             | Future Technology Devices International |
| GPIO             | General Purpose Input/Output            |
| I <sup>2</sup> C | Inter-Integrated Circuit                |
| I <sup>2</sup> S | Inter-IC Sound                          |
| JTAG             | Joint Test Action Group                 |
| LDO              | Low Dropout                             |
| LED              | Light Emitting Diode                    |
| LVDS             | Low-Voltage Differential Signaling      |
| MIPI             | Mobile Industry Processor Interface     |
| ML               | Machine Learning                        |
| MSPI             | Master SPI                              |
| PMOD             | Peripheral Module                       |
| SMA              | SubMiniature version A                  |
| SPI              | Serial Peripheral Interface             |
| SSPI             | Slave SPI                               |
| VIP              | Video Interface Platform                |
| VTT              | Tracking Termination Voltage            |

# 1. Introduction

This document describes the Lattice Semiconductor CrossLinkU™-NX Evaluation Board. The board's key component is the CrossLinkU-NX FPGA (from here on referred as LIFCL-33U), which receives input from a camera and sends video output over LIFCL-33U's USB 3.0 interface.

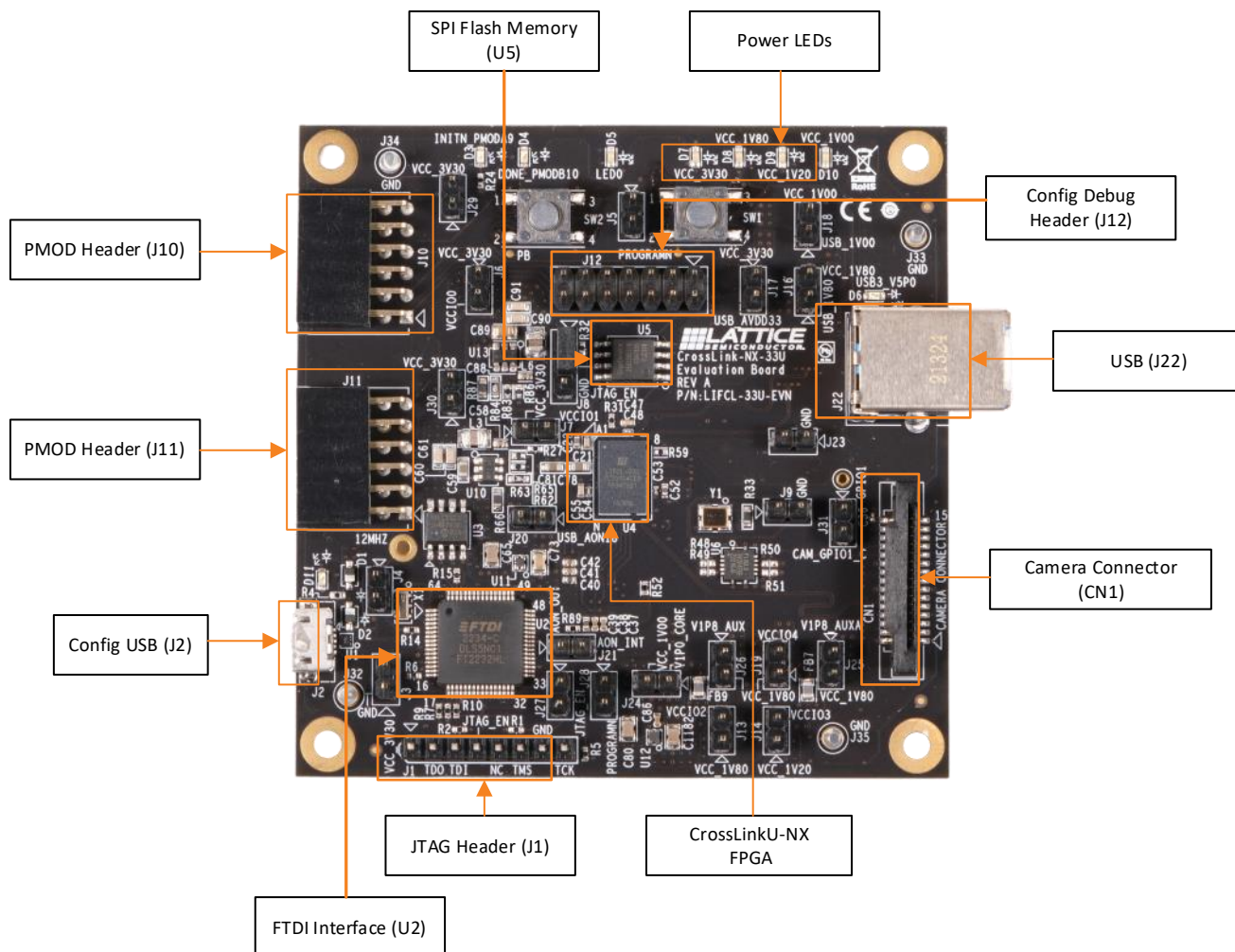
This board also features two PMOD (Peripheral Module) connectors for off board support.

The content of this user guide includes descriptions of onboard settings, connectors, programming circuit, a complete set of schematics, and bill of materials for the CrossLinkU-NX Evaluation Board.

The key features of the CrossLinkU-NX Evaluation Board include:

- LIFCL-33U FPGA (LIFCL-33U-CTG104)
  - MIPI CSI-2 receiver (Soft D-PHY) interface for camera data
  - SPI flash configuration
- Board Resources
  - Two PMOD expansion headers available
  - PSRAM Memory
- Programming Circuit
  - Radiant Programmer through USB/FTDI interface (JTAG or SPI)
  - From onboard Flash
  - External SPI Programmer through header

[Figure 1.1](#) and [Figure 1.2](#) show the top and bottom views of the CrossLinkU-NX Evaluation Board and its key components.



**Figure 1.1. Top View of CrossLinkU-NX Evaluation Board**



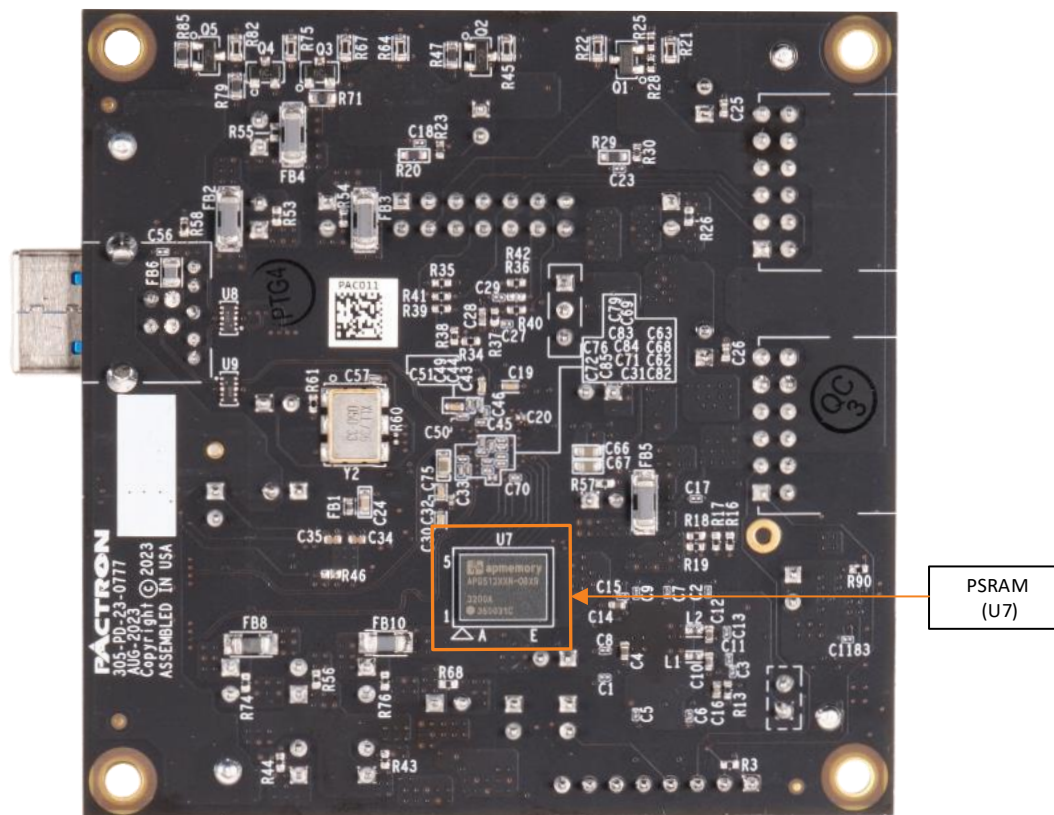


Figure 1.2. Bottom View of CrossLinkU-NX Evaluation Board

## 1.1. Further Information

The following references provide detailed information on the CrossLinkU-NX Evaluation Board and the LIFCL-33U FPGA device:

- [Appendix A. CrossLinkU-NX Evaluation Board Schematics](#)
- [Appendix B. CrossLinkU-NX Evaluation Board Bill of Materials](#)
- [www.latticesemi.com/boards](http://www.latticesemi.com/boards) for more information on boards and kits
- [CrossLink-NX-33 and CrossLinkU-NX Data Sheet \(FPGA-DS-02104\)](#) for details on the LIFCL-33U FPGA.

## 2. Headers and Jumpers

Table 2.1 lists the headers and jumpers as shown in Figure 1.1 and Figure 1.2.

**Table 2.1. Headers and Test Connectors**

| Part | Description                                                 | Settings (Default first)                                                            |
|------|-------------------------------------------------------------|-------------------------------------------------------------------------------------|
| J1   | JTAG Header                                                 | —                                                                                   |
| J3   | FTDI Reset Jumper                                           | Open – Active FTDI<br>Closed – Holds FTDI in reset                                  |
| J4   | Routes 12 MHz clock to Test Point                           | Open – No 12 MHz Clock<br>Closed – 12 MHz Clock to Test Point 12MHz                 |
| J5   | PROGRAMN Pull-down Jumper                                   | Open – MSPI Boot Mode<br>Closed – Secondary Configuration Mode                      |
| J6   | Power Measurement Header for VVCIO0                         | —                                                                                   |
| J7   | Power Measurement Header for VVCIO1                         | —                                                                                   |
| J8   | JTAG_EN - LIFCL-33U (U4)                                    | 1-2 – JTAG is Enabled<br>2-3 – JTAG is Disabled                                     |
| J9   | 25 MHz OSC (Y1) Standby                                     | Open – Y1 Output Active<br>Closed – Y1 in Standby                                   |
| J10  | PMOD Header                                                 | —                                                                                   |
| J11  | PMOD Header                                                 | —                                                                                   |
| J12  | Configuration Debug Header                                  | —                                                                                   |
| J13  | Power Measurement Header for VVCIO2                         | —                                                                                   |
| J14  | Power Measurement Header for VVCIO3                         | —                                                                                   |
| J16  | Power Measurement Header for AVDD18, AVDD18_TX & AVDD18_COM | —                                                                                   |
| J17  | Power Measurement Header for AVDD33                         | —                                                                                   |
| J18  | Power Measurement Header for AVDD & AVDD_TX                 | —                                                                                   |
| J19  | Power Measurement Header for VVCIO4                         | —                                                                                   |
| J20  | Power Measurement Header for VCCAUX_AON                     | —                                                                                   |
| J21  | AON Header                                                  | —                                                                                   |
| J23  | 50 MHz OSC (Y2) Enable/Disable                              | Open – Enable<br>Closed - Disable                                                   |
| J24  | Power Measurement Header for VCC                            | —                                                                                   |
| J25  | Power Measurement Header for VCCAUX (H7)                    | —                                                                                   |
| J26  | Power Measurement Header for VCCAUX (Other Balls)           | —                                                                                   |
| J27  | FTDI JTAG_EN Connection Header                              | Open – JTAG_EN not connected to FTDI<br>Closed – JTAG_EN connected to FTDI          |
| J28  | FTDI PROGRAMN Connection Header                             | Open – PROGRAMN not connected to FTDI<br>Closed – PROGRAMN connected to FTDI        |
| J29  | Power Connection for PMOD J10                               | Closed – 3.3V connected to PMOD Header<br>Open – Power not connected to PMOD Header |
| J30  | Power Connection for PMOD J11                               | Closed – 3.3V connected to PMOD Header<br>Open – Power not connected to PMOD Header |
| J31  | Connection to Test Point GPIO1                              | Open – Ball L6 connected to CN1 Connector<br>Closed – Ball L6 connected to TP GPIO1 |

## 3. Board Programming

### 3.1. Programming Circuit

The LIFCL-33U can be programmed with USB through the FTDI/JTAG interface using Lattice Radiant™ programmer software, or by an external programmer connected to Header J9.

Figure 3.1 shows the programming block of CrossLinkU-NX Evaluation Board.

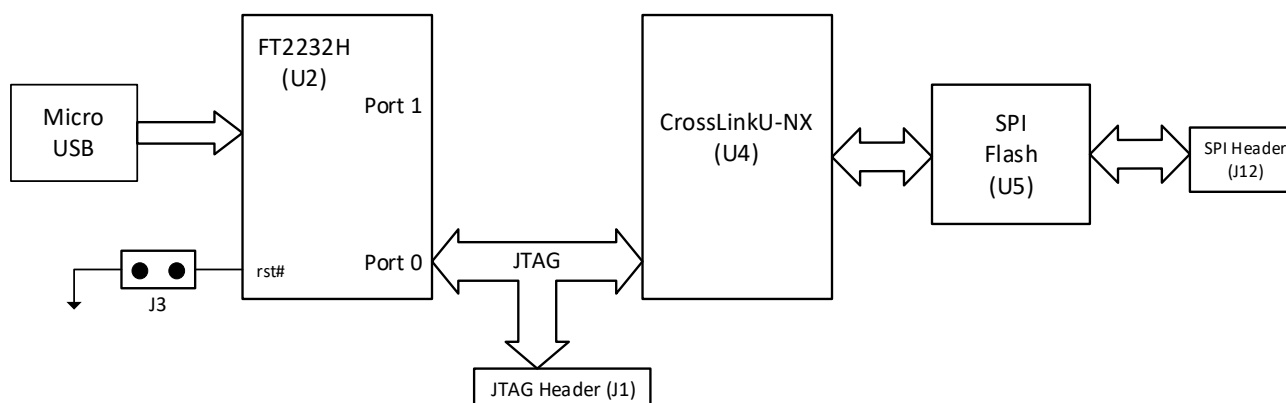


Figure 3.1. Programming Block

The FTDI/JTAG interface is used to program both LIFCL-33U and SPI Flash Memory (GigaDevice GD25Q128E 128 Mb).

### 3.2. Programming the Board - SRAM

This section describes the procedure for programming a pattern to the SRAM (volatile) configuration memory of LIFCL-33U. The LIFCL-33U can be programmed through the JTAG, I<sup>2</sup>C, I<sup>3</sup>C or SPI interfaces. This section focuses on JTAG programming through the USB/FTDI interface. For details on the other configuration modes, refer to the [sysCONFIG Usage Guide for Nexus Platform \(FPGA-TN-02099\)](#).

The board is programmed using the Lattice Radiant Programmer software, which can be started as a stand-alone tool or from a Lattice Radiant project.

To program the board:

1. Power ON the board by connecting the USB cable to J2 of CrossLinkU-NX Evaluation Board.
2. Start a programming project by launching the tool and initiating a board scan, as shown in Figure 3.2.

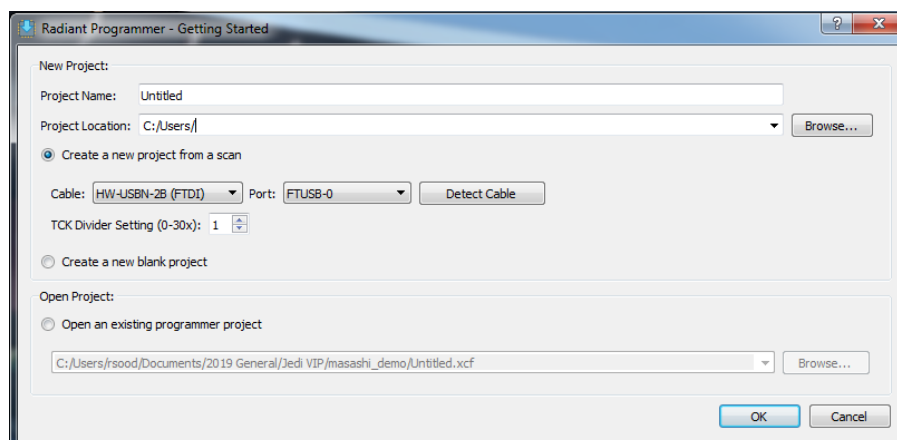
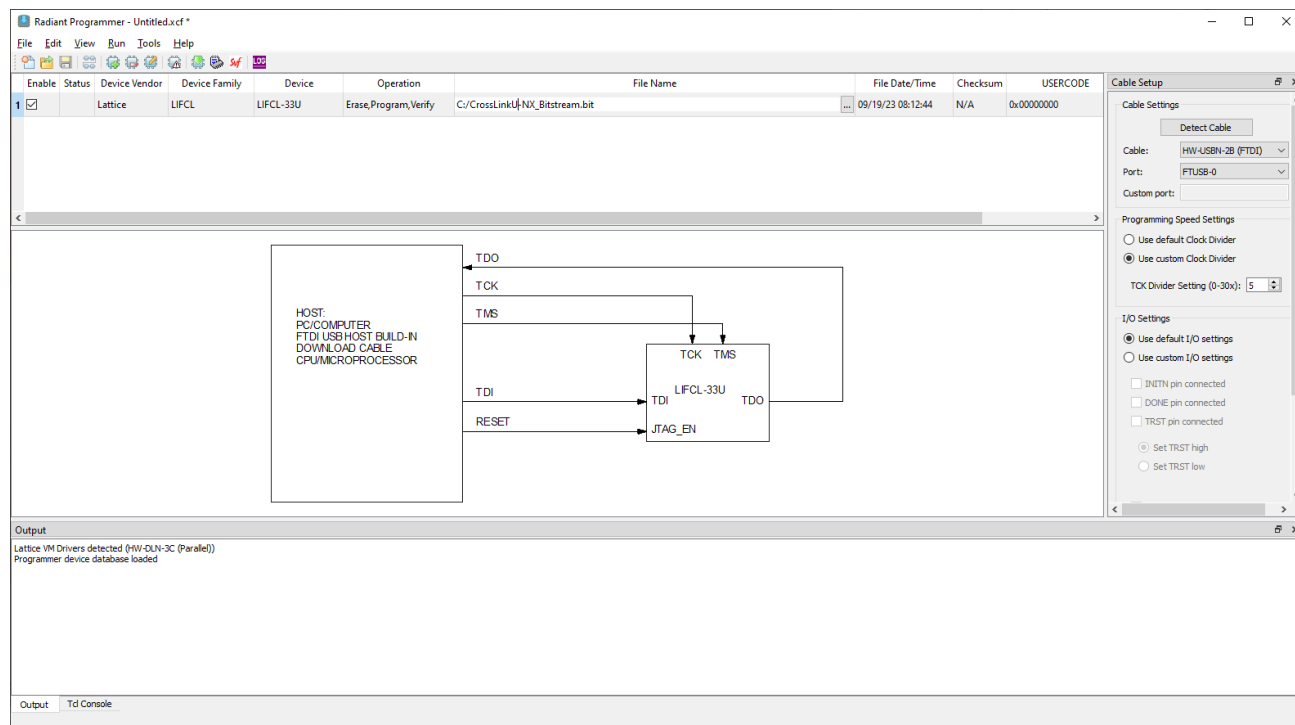


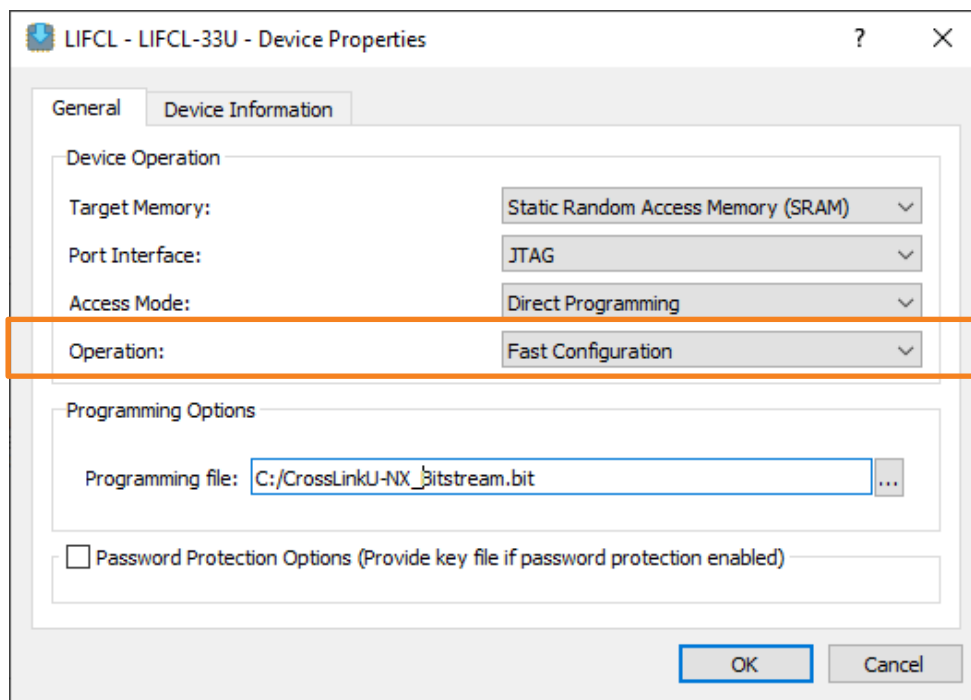
Figure 3.2. Starting Programmer

- When the board is successfully scanned, the window shown in [Figure 3.3](#) opens. This interface allows you to enter the file name.



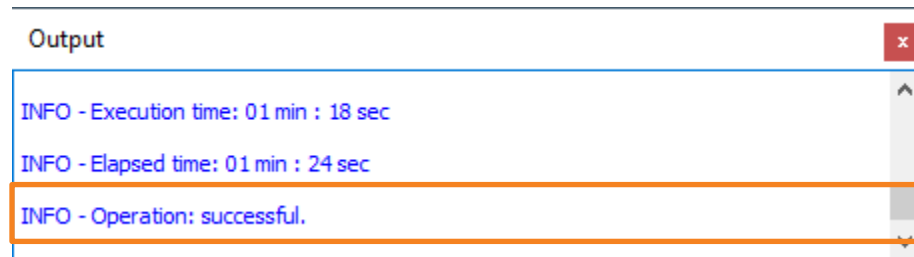
**Figure 3.3. Lattice Radiant Programmer – Fast Configuration**

- Double-click on the **Operation** field and select the appropriate programming mode. In this example, **Fast Configuration** of the SRAM array through JTAG is selected.



**Figure 3.4. Entering Programming Mode**

5. Click the **Program** button to configure the LIFCL-33U on the board.
6. Check the **Output** console at the bottom of Lattice Radiant Programmer for the result of the programming. You should see **Operation: Successful** as shown in [Figure 3.5](#).

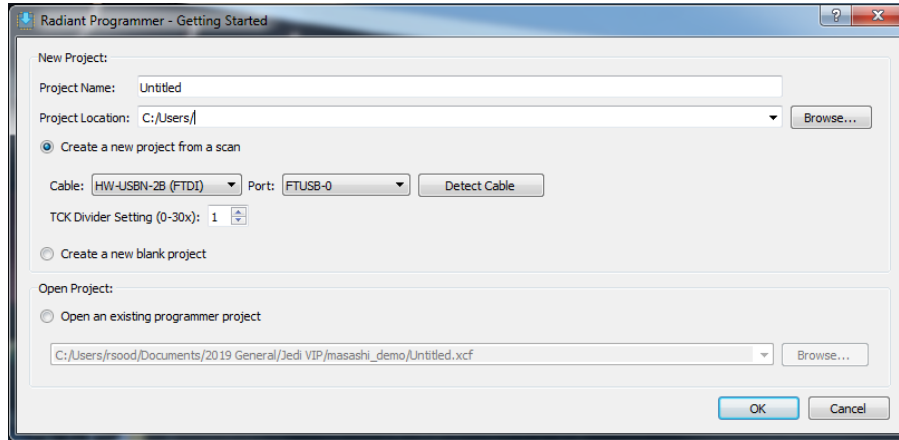


**Figure 3.5. Lattice Radiant Programmer Output Window**

### 3.3. Programming the Board – Flash

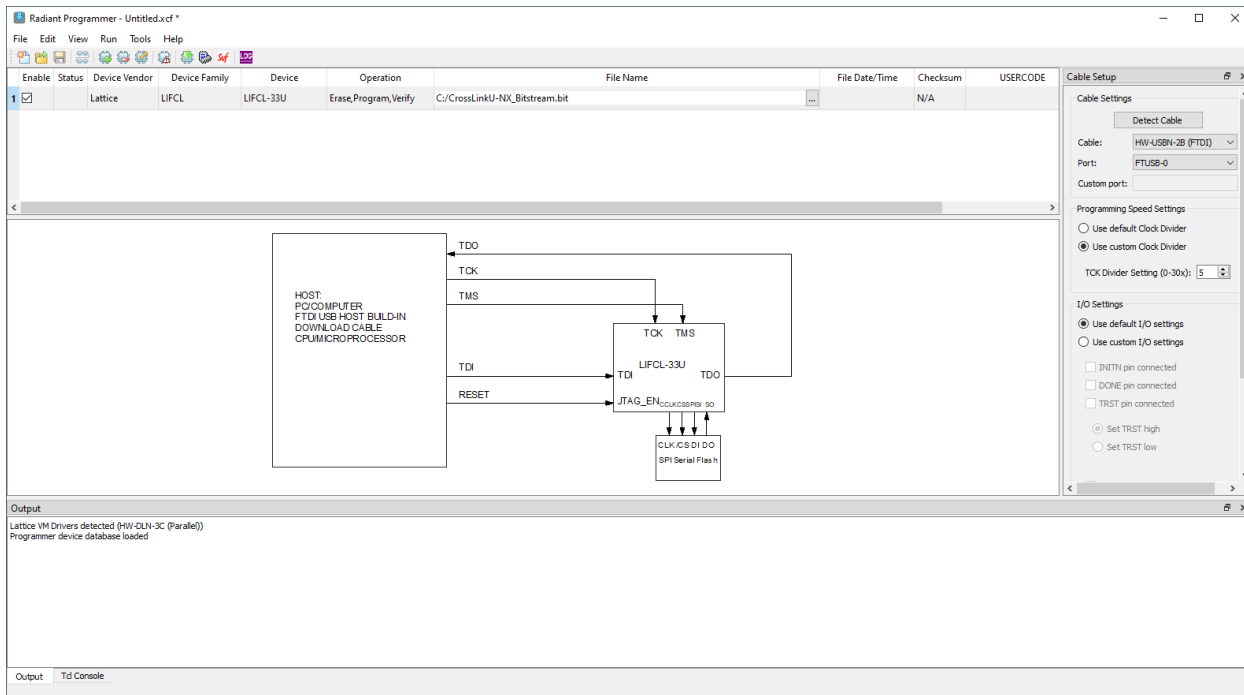
To program the on board Flash:

1. Power ON the board by connecting the USB cable to J2 of the LIFCL-33U-EVN Board.
2. Start a programming project by launching the tool and initiating a board scan, as shown in [Figure 3.6](#).



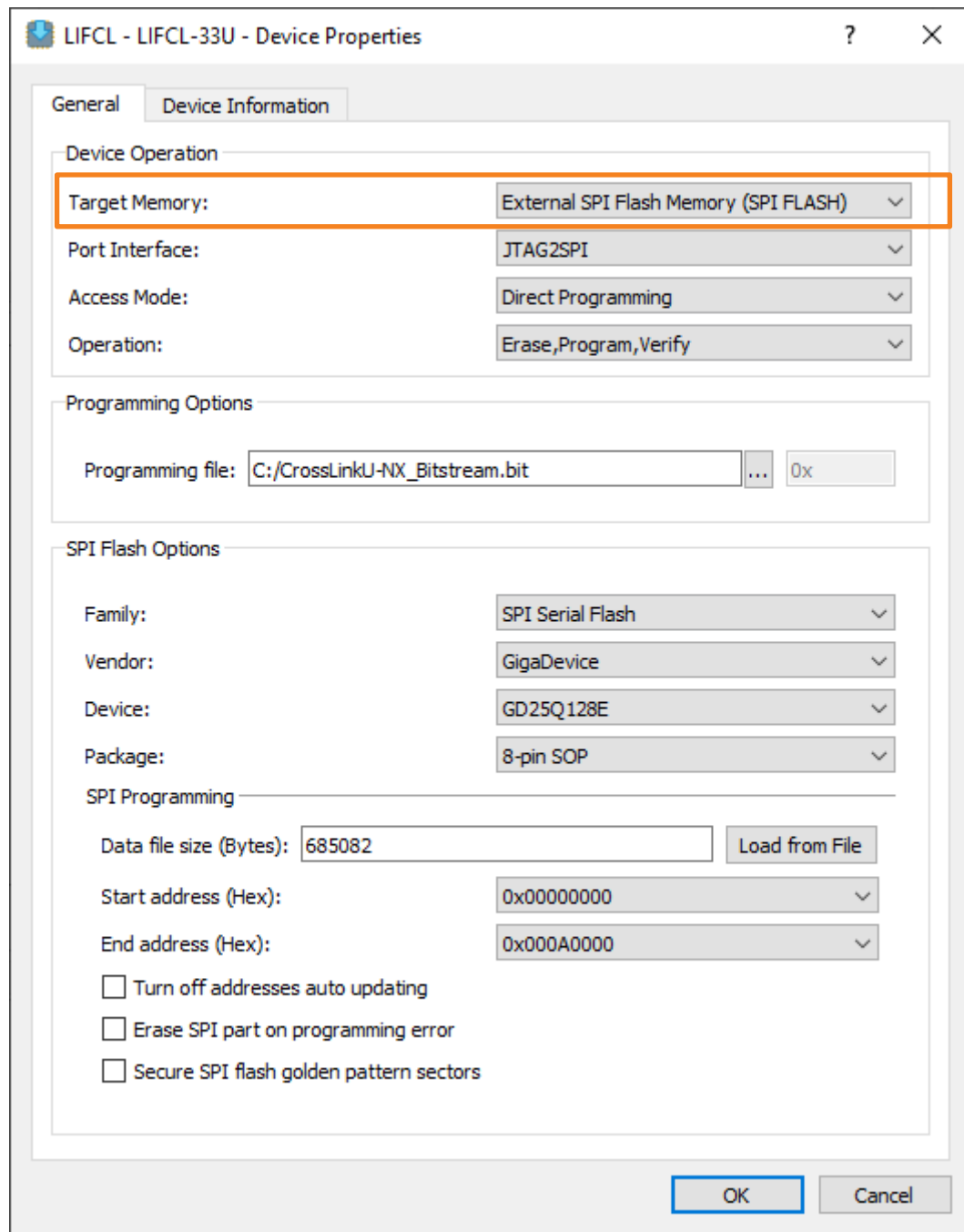
**Figure 3.6. Starting Programmer**

3. When the board is successfully scanned, a window opens as shown in [Figure 3.7](#). This interface allows you to enter the file name.



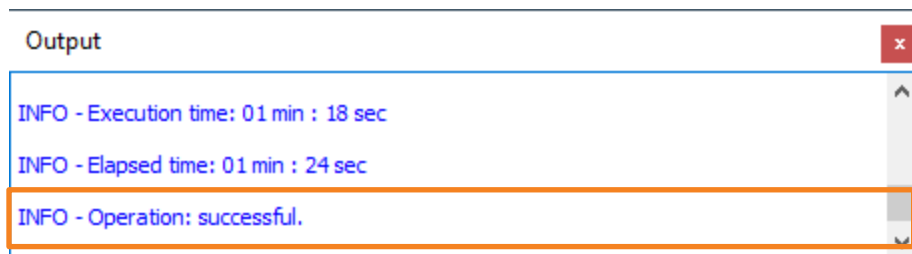
**Figure 3.7. Lattice Radiant Programmer – External SPI Flash Memory**

4. Double-click on the **Operation** field and **External SPI Flash Memory (SPI FLASH)**.



**Figure 3.8. Entering Programming Mode**

5. Click **OK** button in the **Device Properties** Window after selecting the appropriate fields
6. Click the **Program** button to configure the Flash device on the board.
7. Check the **Output** console at the bottom of the Lattice Radiant Programmer for the result of the programming. You should see **Operation: Successful** as shown in [Figure 3.9](#).



**Figure 3.9. Lattice Radiant Programmer Output Window**



## 4. CrossLinkU-NX Interface Support

The CrossLinkU-NX Evaluation Board supports various onboard interfaces and external interfaces. The sections below describe key onboard interfaces supported on the CrossLinkU-NX Evaluation Board.

### 4.1. Camera Sensor Interface

Figure 4.1 shows the block diagram of the camera sensor interface. The CN1 connector supports up to two MIPI data lanes. The camera sensor can be configured using the I<sup>2</sup>C interface from the LIFCL-33U. CN1 connector is compatible with Raspberry Pi camera modules.

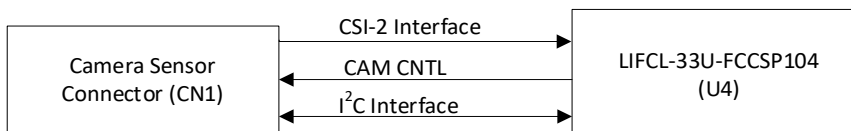


Figure 4.1. Camera Sensor Interface

### 4.2. PMOD Interfaces

Figure 4.2 shows the block diagram of the PMOD interfaces. The LIFCL-33U is connected to two PMOD connectors, J10 and J11. Power to the PMOD connectors can be controlled independently through J29 for J10 and J30 for J11.

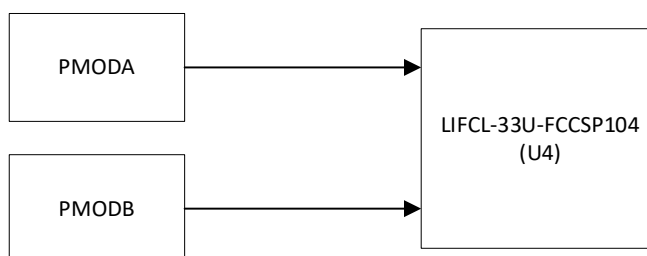
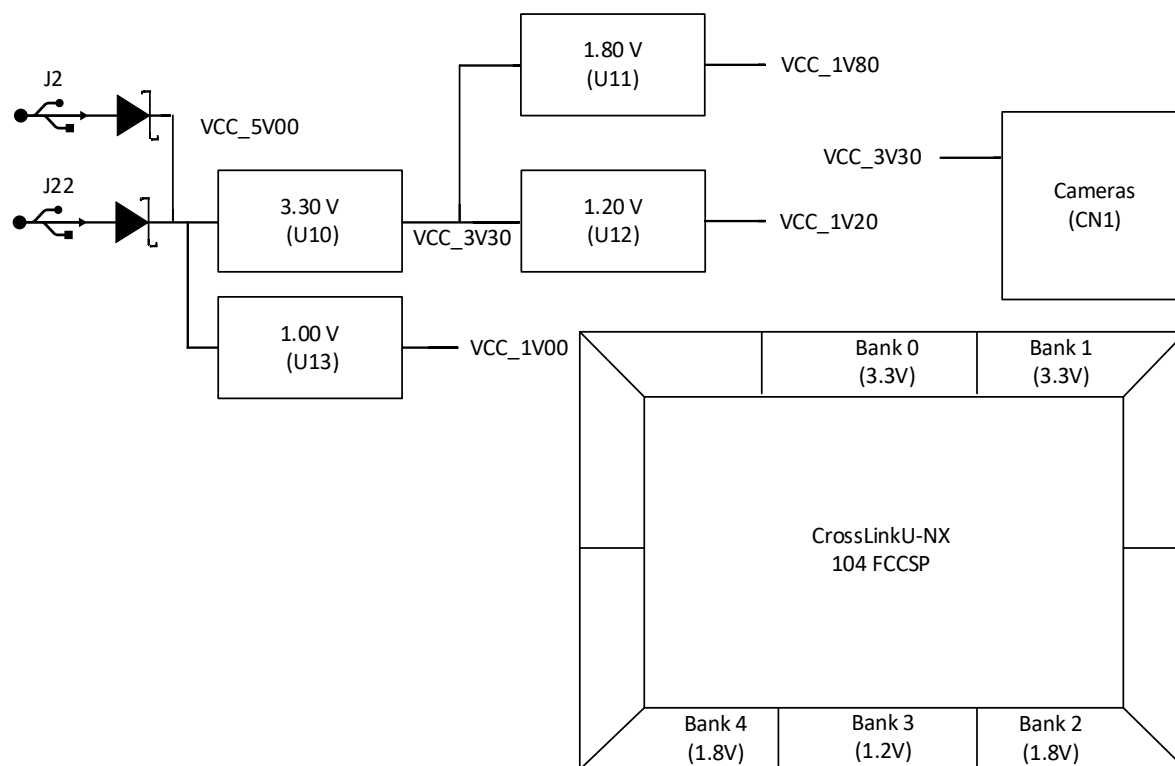


Figure 4.2. PMOD Interfaces

## 5. Power Supply

The power supply to the CrossLinkU-NX Evaluation Board is provided from the J2 or J22 USB connectors. If either or both USB ports are connected to a powered cable, the board will power up.

Figure 5.1 shows the power supply block of the CrossLinkU-NX Evaluation Board. The 5 V is supplied through the USB, and then board converters create the remaining 3.3 V, 1.8 V, 1.2 V, and 1.0 V supplies required by the board components.



**Figure 5.1. Power Supply Block**

Table 5.1 lists the board voltage rails, including rail source voltage, voltage on net, LED, and LED color.

**Table 5.1. Device Power Rail Summary**

| Voltage Rail | Source Rail | Voltage on Net (V) | Status LED | LED Color |
|--------------|-------------|--------------------|------------|-----------|
| VCC_5V00     | USB         | 5.00               | N/A        | N/A       |
| VCC_3V30     | VCC_5V00    | 3.30               | D7         | Green     |
| VCC_1V80     | VCC_3V30    | 1.80               | D8         | Green     |
| VCC_1V20     | VCC_3V30    | 1.20               | D9         | Green     |
| VCC_1V00     | VCC_5V00    | 1.00               | D10        | Green     |

## 6. CrossLinkU-NX I/O Ball Mapping to Connectors

**Table 6.1. Camera Sensor Connector Pin Mapping**

| CN1     |                     |          |
|---------|---------------------|----------|
| Pin Num | Net/Test Point Name | Ball Num |
| 1       | GND                 | —        |
| 2       | CAM_D0_N            | M6       |
| 3       | CAM_D0_P            | M7       |
| 4       | GND                 | —        |
| 5       | CAM_D1_N            | N6       |
| 6       | CAM_D1_P            | N7       |
| 7       | GND                 | —        |
| 8       | CAM_CK_N            | E6       |
| 9       | CAM_CK_P            | F7       |
| 10      | GND                 | —        |
| 11      | CAM_GPIO0           | F7       |
| 12      | CAM_GPIO1_C         | L6       |
| 13      | CAM_SCL             | J6       |
| 14      | CAM_SDA             | H6       |
| 15      | VCC_3V30            | —        |

**Table 6.2. JTAG Connector Pin Mapping**

| J1      |                     |          |
|---------|---------------------|----------|
| Pin Num | Net/Test Point Name | Ball Num |
| 1       | VCC_3V30            | —        |
| 2       | FTDI_TDO_PMODB1     | D1       |
| 3       | FTDI_TDI_PMODB2     | D2       |
| 4       | JTAG_EN             | A2       |
| 5       | NC                  | —        |
| 6       | FTDI_TMS_PUSHBUTTON | B2       |
| 7       | GND                 | —        |
| 8       | FTDI_TCK_PMODB8     | F2       |

Table 6.3. PMOD Connector Pin Mapping

| PMODA (J10) |                 |             |
|-------------|-----------------|-------------|
| Pin Number  | Net Name        | Ball Number |
| 1           | SPI_MCLK_PMODA1 | B4          |
| 2           | CSSPIN_PMODA2   | B3          |
| 3           | DQ0_MOSI_PMODA3 | D4          |
| 4           | DQ1_MISO_PMODA4 | D3          |
| 5           | GND             | —           |
| 6           | VCC_3V30        | —           |
| 7           | DQ2_PMODA7      | E4          |
| 8           | DQ3_PMODA8      | E3          |
| 9           | INITN_PMODA9    | A1          |
| 10          | DONE_PMODA10    | G2          |
| 11          | GND             | —           |
| 12          | VCC_3V30        | —           |

| PMODB (J11) |                 |             |
|-------------|-----------------|-------------|
| Pin Number  | Net Name        | Ball Number |
| 1           | FTDI_TDO_PMODB1 | D1          |
| 2           | FTDI_TDI_PMODB2 | D2          |
| 3           | PMODB3          | E1          |
| 4           | PMODB4          | E2          |
| 5           | GND             | —           |
| 6           | VCC_3V30        | —           |
| 7           | PMODB7          | F1          |
| 8           | FTDI_TCK_PMODB8 | F2          |
| 9           | PMODB9          | G1          |
| 10          | PMODA10         | B1          |
| 11          | GND             | —           |
| 12          | VCC_3V30        | —           |

**Table 6.4. Configuration Debug Connector Pin Mapping**

| J12        |                 |             |
|------------|-----------------|-------------|
| Pin Number | Net Name        | Ball Number |
| 1          | PROGRAMN        | A4          |
| 2          | NC              | —           |
| 3          | CSSPIN_PMODA2   | B3          |
| 4          | DONE_PMODB10    | G2          |
| 5          | DQ0_MOSI_PMODA3 | D4          |
| 6          | INITN_PMODA9    | A1          |
| 7          | DQ1_MISO_PMODA4 | D3          |
| 8          | NC              | —           |
| 9          | DQ2_PMODA7      | E4          |
| 10         | VCC_3V30        | —           |
| 11         | DQ3_PMODA8      | E3          |
| 12         | SPI_MCLK_PMODA1 | B4          |
| 13         | GND             | —           |
| 14         | GND             | —           |

## 7. Status Indicators

The LED status indicators on the board show the application status. [Table 7.1](#) lists the status LED I/O map.

**Table 7.1. Status LED I/O Map**

| Net Name | LED | Connector/Pin | Color |
|----------|-----|---------------|-------|
| LED0     | D5  | G6            | Green |

## 8. Pushbuttons

[Table 8.1](#) lists all the pushbuttons. This board has two pushbuttons.

**Table 8.1. Switch and Pushbutton I/O Map**

| Net Name            | Component | Connector/Pin |
|---------------------|-----------|---------------|
| PROGRAMN            | SW1       | A4            |
| FTDI_TMS_PUSHBUTTON | SW2       | B2            |

## 9. Ordering Information

This board is included as part of a kit, and not available as a separate item. The part number in [Table 9.1](#) is for reference only, so it is clear which board is described in this document. Visit [www.latticesemi.com/boards](http://www.latticesemi.com/boards) for the latest ordering information.

**Table 9.1. Reference Part Number**

| Description                    | Ordering Part Number |
|--------------------------------|----------------------|
| CrossLinkU-NX Evaluation Board | LIFCL-33U-EVN        |





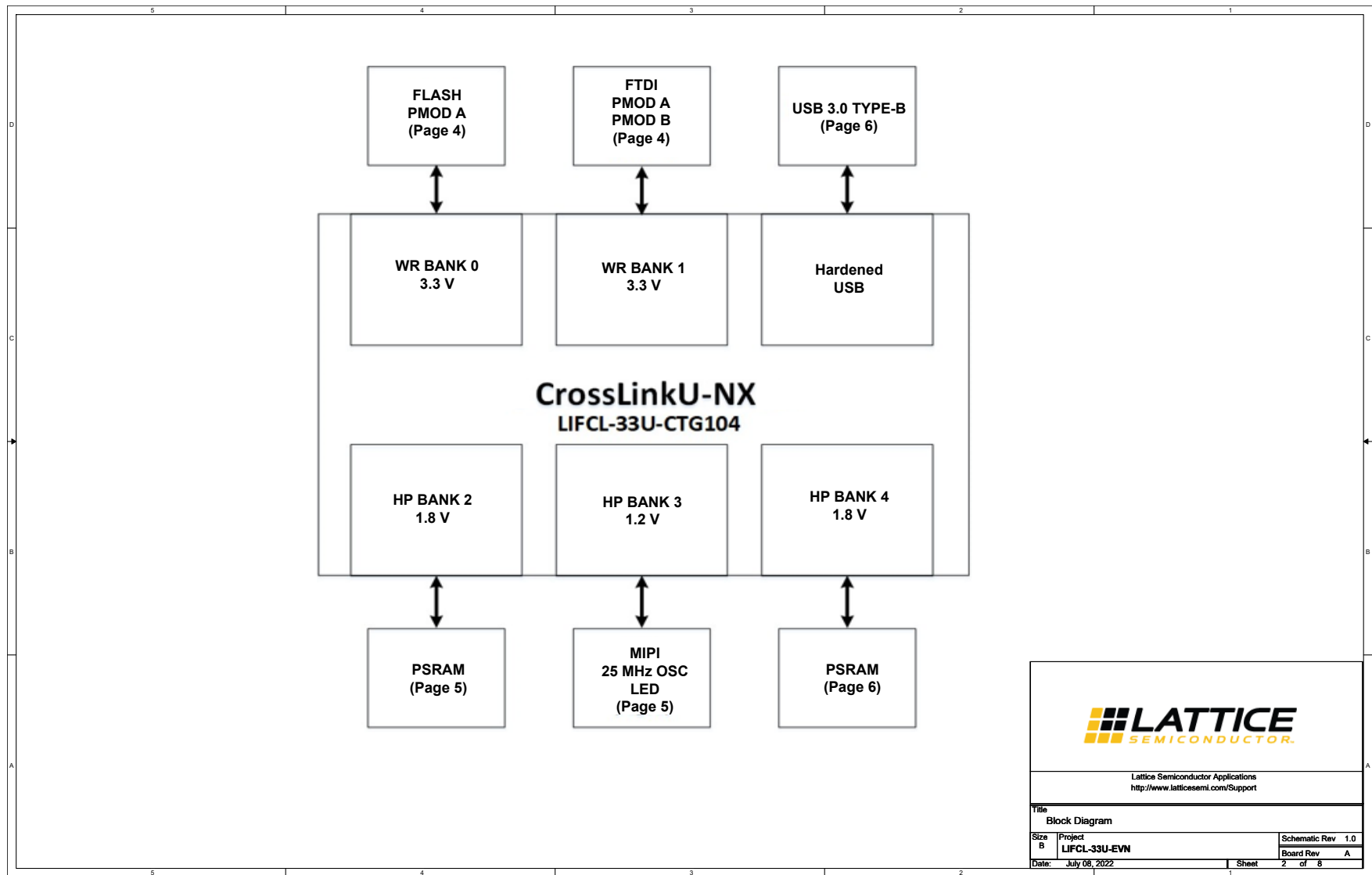


Figure A.2. Block Diagram

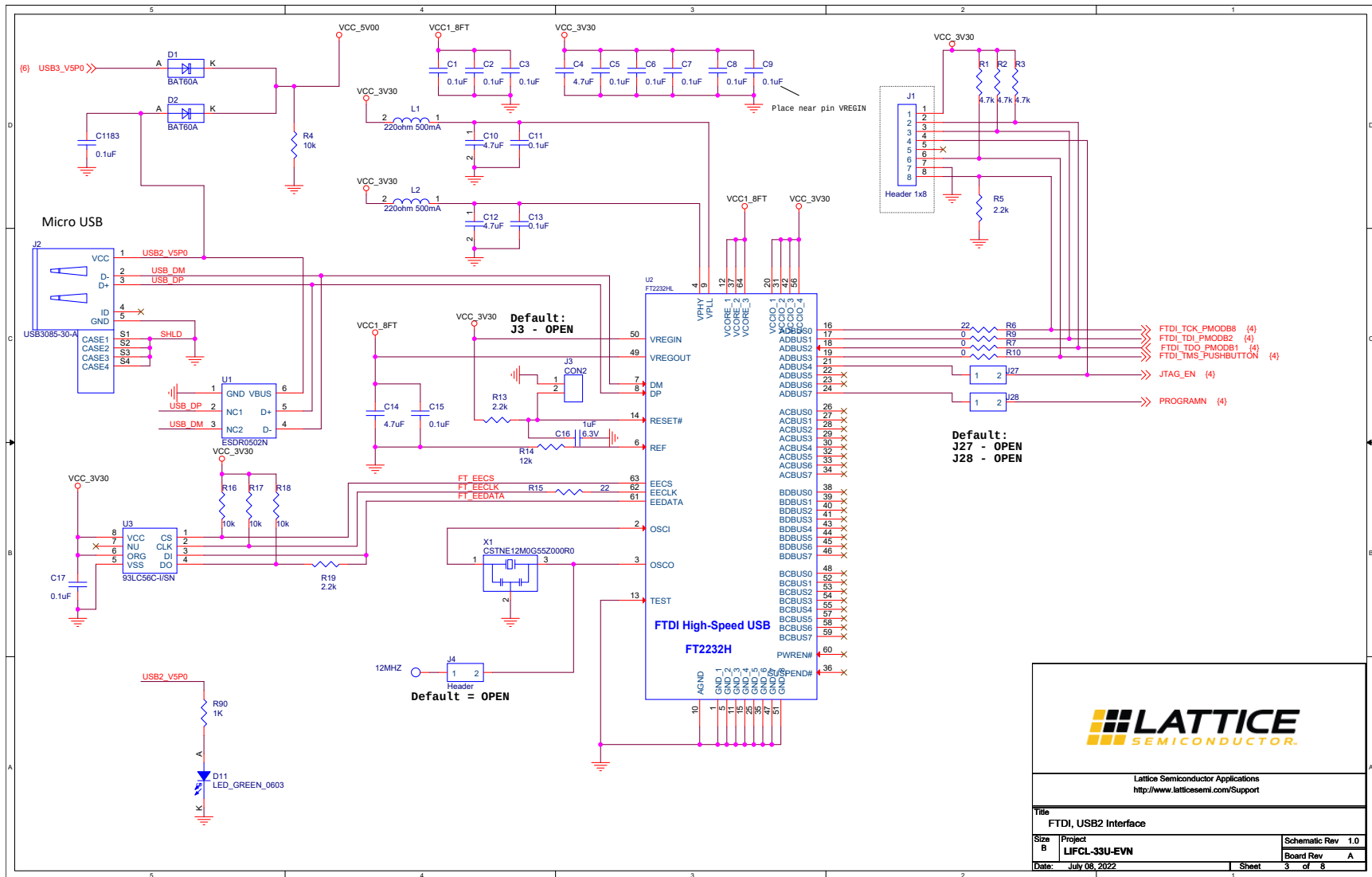


Figure A.3. FTDI, USB2 Interface

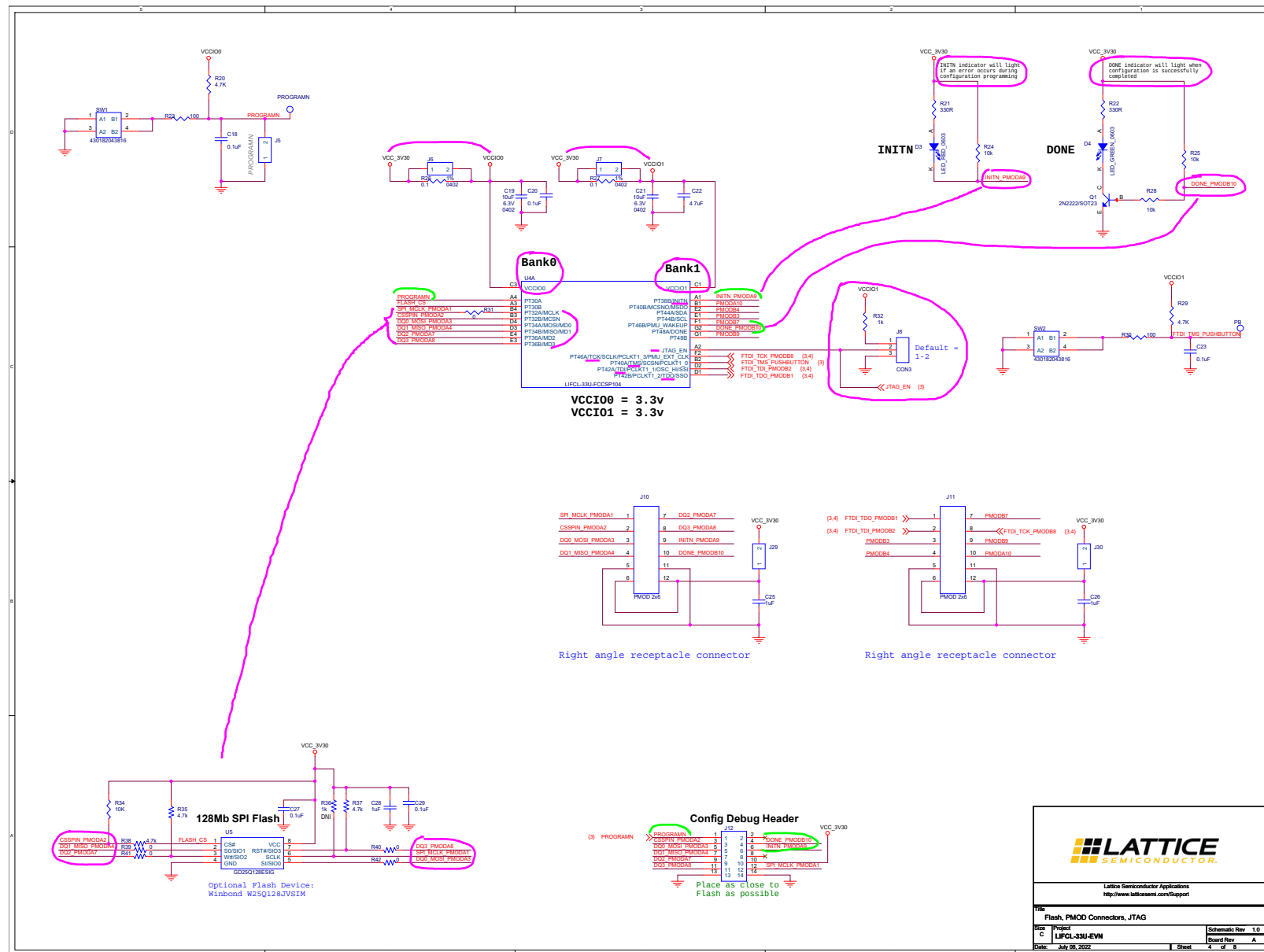


Figure A.4. Flash, PMOD Connectors, JTAG



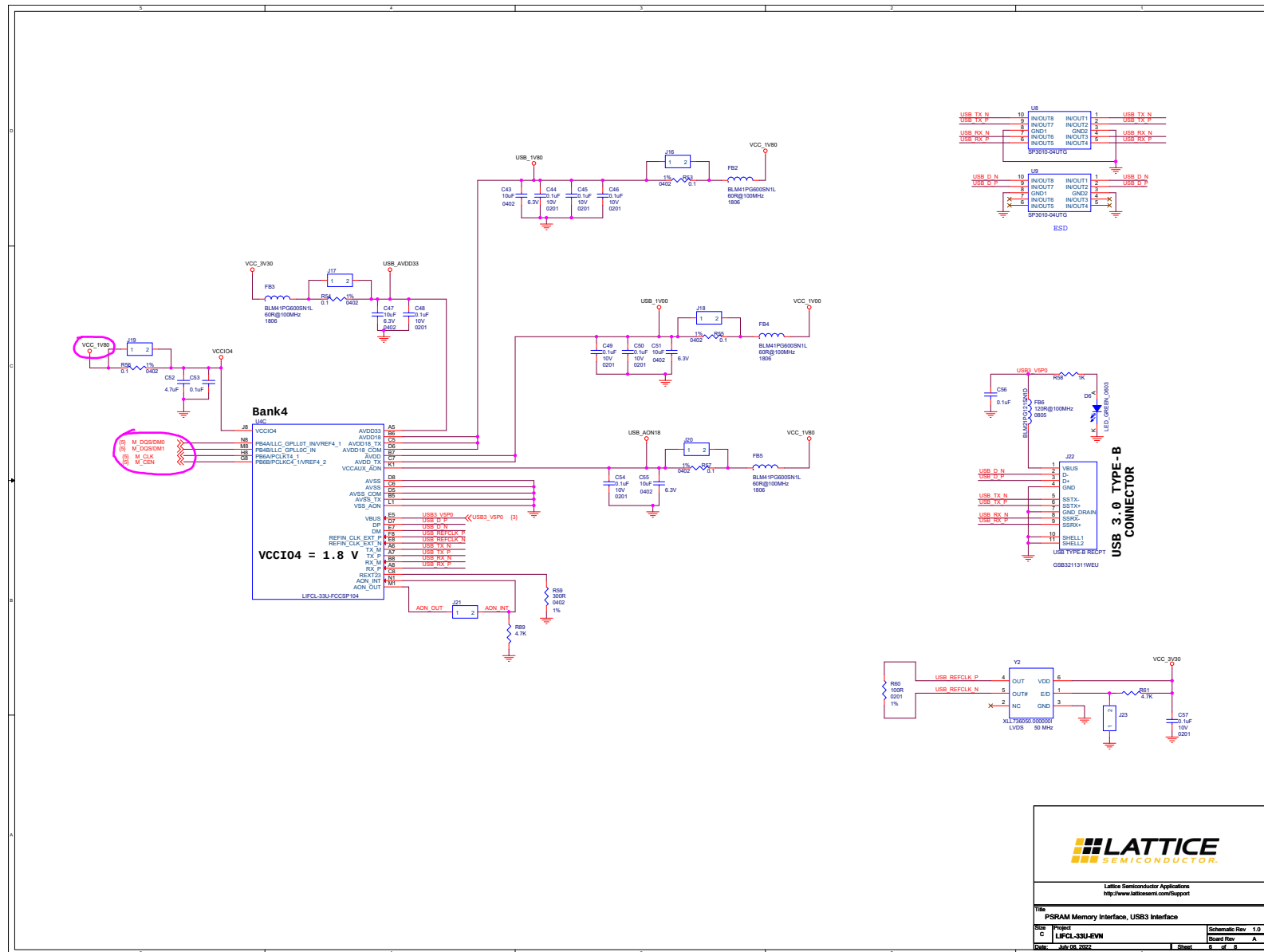


Figure A.6. PSRAM Memory Interface, USB 3 Interface

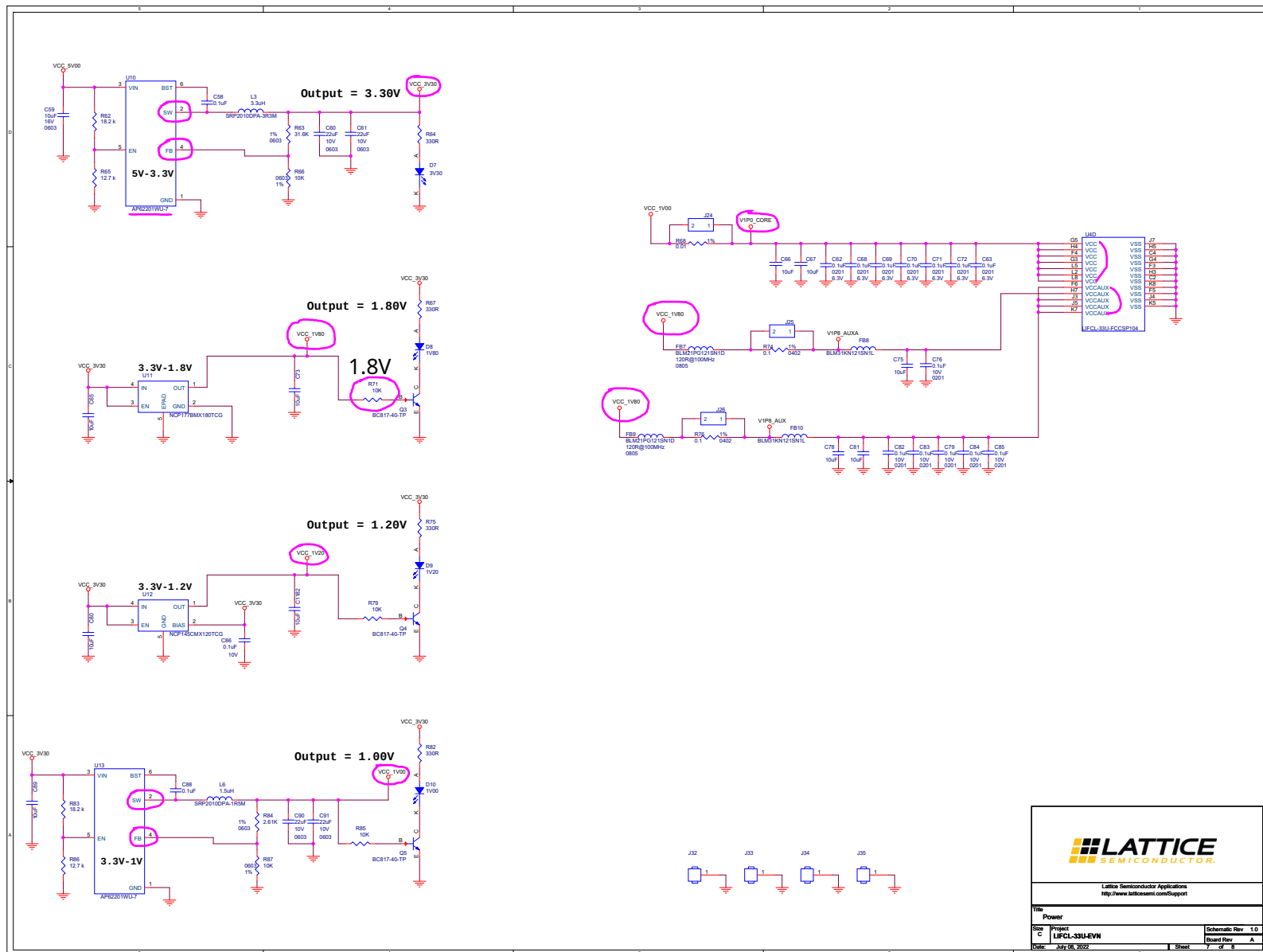
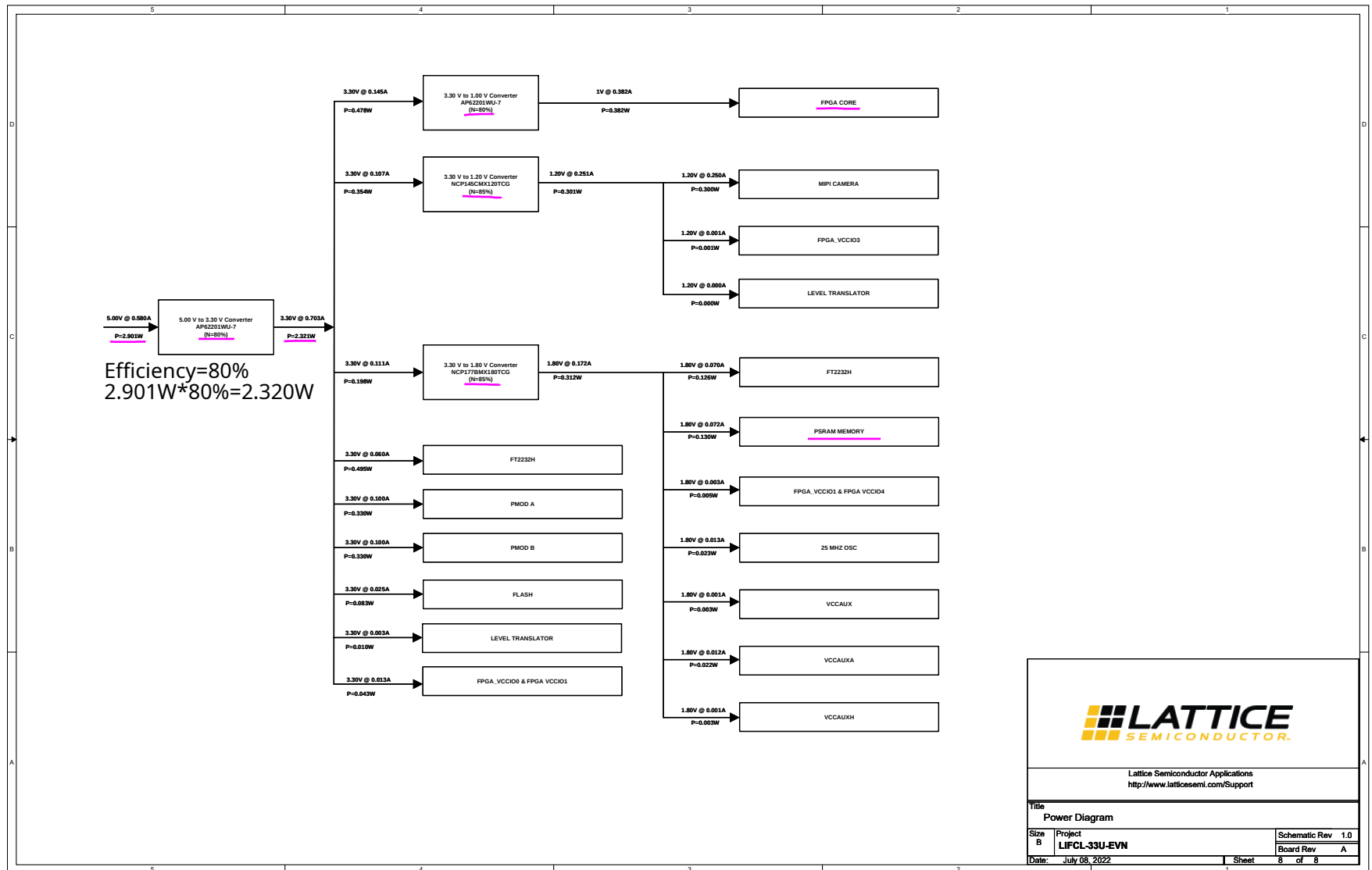


Figure A.7. Power



**Figure A.8. Power Diagram**

## Appendix B. CrossLinkU-NX Evaluation Board Bill of Materials

| Item | Reference                                               | Qty | Value       | PCB Footprint | Comments | Manufacturer Part Number | Manufacturer                         | Description                                |
|------|---------------------------------------------------------|-----|-------------|---------------|----------|--------------------------|--------------------------------------|--------------------------------------------|
| 1    | CN1                                                     | 1   | 1-1734248-5 | 15P_Connector | -        | 1-1734248-5              | TE Connectivity<br>AMP<br>Connectors | CONN FPC VERT 15POS 1MM<br>SMD             |
| 2    | C1,C2,C3,C5,C6,C7,C8,C9,C11,C13,C15,C17,C27,C29         | 14  | 0.1uF       | C0201         | -        | C0603X5R1C104K030BC      | TDK                                  | CAP CER 0.1UF 16V X5R 0201                 |
| 3    | C4,C10,C12,C14,C22,C39,C40,C52                          | 8   | 4.7uF       | C0402         | -        | C1005X5R1A475K050BC      | TDK                                  | CAP CER 4.7UF 10V X5R 0402                 |
| 4    | C16,C25,C26,C28                                         | 4   | 1uF         | C0402         | -        | EMK105BJ105KVHF          | Taiyo Yuden                          | CAP CER 1UF 16V X5R 0402                   |
| 5    | C18,C23                                                 | 2   | 0.1uF       | C0201         | -        | C0603X5R1C104K030BC      | TDK                                  | CAP CER 0.1UF 16V X5R 0201                 |
| 6    | C19,C21,C30,C32                                         | 4   | 10uF        | 402           | -        | GRM155R60J106ME15D       | Murata<br>Electronics                | CAP CER 10UF 6.3V X5R 0402                 |
| 7    | C20,C31,C33,C36,C53                                     | 5   | 0.1uF       | c0201_chamfer | -        | C0603X5R1C104K030BC      | TDK                                  | CAP CER 0.1UF 16V X5R 0201                 |
| 8    | C24                                                     | 1   | 1uF         | C0603         | -        | CL10B105KP8NNNC          | Samsung<br>Electro-<br>Mechanics     | CAP CER 1UF 10V X7R 0603                   |
| 9    | C34,C35,C37,C38,C41,C42,C86                             | 7   | 0.1uF       | C0402         | -        | GRM155R61A104KA01J       | Murata<br>Electronics                | CAP CER 0.1UF 10V X5R 0402                 |
| 10   | C43,C47,C51,C55                                         | 4   | 10uF        | C0402         | -        | GRM155R60J106ME15D       | Murata<br>Electronics                | CAP CER 10UF 6.3V X5R 0402                 |
| 11   | C44,C45,C46,C48,C49,C50,C54,C57,C76,C79,C82,C83,C84,C85 | 14  | 0.1uF       | C0201         | -        | GRM033R61A104ME15D       | Murata                               | 0.1uF Ceramic Chip<br>Capacitor, X5R, 10V  |
| 12   | C56,C1183                                               | 2   | 0.1uF       | C0201         | -        | GRM033R61E104KE14J       | Murata                               | CAP CER 0.1UF 25V 10% X5R<br>0201          |
| 13   | C58,C88                                                 | 2   | 0.1uF       | cap0402       | -        | GRM155R71H104KE14J       | Murata<br>Electronics                | CAP CER 0.1UF 50V X7R 0402                 |
| 14   | C59                                                     | 1   | 10uF        | C0603         | -        | EMK107BBJ106MA-T         | Taiyo Yuden                          | CAP CER 10UF 16V X5R 0603                  |
| 15   | C60,C61,C90,C91                                         | 4   | 22uF        | C0603         | -        | CL10A226MP8NUNE          | Samsung<br>Electro-<br>Mechanics     | CAP CER 22UF 10V X5R 0603                  |
| 16   | C62,C63                                                 | 2   | 0.1uF       | C0201         | -        | GRM033R60J104KE19D       | Murata                               | 0.1uF Ceramic Chip<br>Capacitor, X5R, 6.3V |
| 17   | C65,C73,C80,C89,C1182                                   | 5   | 10uF        | cap0805       | -        | C2012X5R1E106M085AC      | TDK<br>Corporation                   | CAP CER 10UF 25V X5R 0805                  |
| 18   | C66,C67,C75,C78,C81                                     | 5   | 10uF        | C0603         | -        | C1608X5R1C106M080AB      | TDK<br>Corporation                   | CAP CER 10UF 16V X5R 0603                  |



| Item | Reference                                                                          | Qty | Value            | PCB Footprint           | Comments              | Manufacturer Part Number | Manufacturer          | Description                                                                       |
|------|------------------------------------------------------------------------------------|-----|------------------|-------------------------|-----------------------|--------------------------|-----------------------|-----------------------------------------------------------------------------------|
| 19   | C68,C69,C70,C71,C72                                                                | 5   | 0.1uF            | C0201                   | -                     | GRM033R60J104KE19D       | Murata Electronics    | 0.1uF Ceramic Chip Capacitor, X5R, 6.3V                                           |
| 20   | D1,D2                                                                              | 2   | BAT60A           | SOD-323                 | -                     | BAT60A E6327             | Infineon Technologies | SCHOTTY DIODE BAT60A                                                              |
| 21   | D3                                                                                 | 1   | LED_RED_0603     | APT1608                 | -                     | APT1608SURCK             | Kingbright            | LED RED CLEAR CHIP SMD                                                            |
| 22   | D4,D6,D11                                                                          | 3   | LED_GREEN_0603   | APT1608                 | -                     | APT1608CGCK              | Kingbright            | LED GREEN CLEAR CHIP SMD                                                          |
| 23   | D5                                                                                 | 1   | LED0             | led_0603                | -                     | SML-D12M8WT86            | Rohm Semiconductor    | LED GREEN DIFFUSED 0603 SMD                                                       |
| 24   | D7                                                                                 | 1   | 3V30             | led_0603                | -                     | SML-D12M8WT86            | Rohm Semiconductor    | LED GREEN DIFFUSED 0603 SMD                                                       |
| 25   | D8                                                                                 | 1   | 1V80             | led_0603                | -                     | SML-D12M8WT86            | Rohm Semiconductor    | LED GREEN DIFFUSED 0603 SMD                                                       |
| 26   | D9                                                                                 | 1   | 1V20             | led_0603                | -                     | SML-D12M8WT86            | Rohm Semiconductor    | LED GREEN DIFFUSED 0603 SMD                                                       |
| 27   | D10                                                                                | 1   | 1V00             | led_0603                | -                     | SML-D12M8WT86            | Rohm Semiconductor    | LED GREEN DIFFUSED 0603 SMD                                                       |
| 28   | FB1                                                                                | 1   | MPZ1005S121CT000 | FB0402                  | -                     | MPZ1005S121CT000         | TDK Corporation       | FERRITE BEAD 120 OHM 0402 1LN                                                     |
| 29   | FB2,FB3,FB4,FB5                                                                    | 4   | 60R@100MHz       | 1806                    | -                     | BLM41PG600SN1L           | Murata Electronics    | FERRITE BEAD 60 OHM 1806 1LN                                                      |
| 30   | FB6,FB7,FB9                                                                        | 3   | 120R@100MHz      | 805                     | -                     | BLM21PG121SN1D           | Murata Electronics    | FERRITE BEAD 120 OHM 0805 1LN                                                     |
| 31   | FB8,FB10                                                                           | 2   | BLM31KN121SN1L   | BLM41P                  | -                     | BLM31KN121SN1L           | Murata Electronics    | FERRITE BEAD 120 OHM 1206 1LN                                                     |
| 32   | GPIO1                                                                              | 1   | TestPoint        | TP50                    | DNL                   | -                        |                       |                                                                                   |
| 33   | J1                                                                                 | 1   | Header 1x8       | hdr_amp_87220_8_1x8_100 | -                     | 22284081                 | Molex                 | CONN HEADER 8POS .100 VERT TIN                                                    |
| 34   | J2                                                                                 | 1   | USB3085-30-A     | MICRO_AB_USB3085_30_A   | -                     | USB3085-30-A             | GCT                   | USB - micro AB USB 2.0 Receptacle Connector 5 Position Surface Mount, Right Angle |
| 35   | J3                                                                                 | 1   | CON2             | CON2                    | -                     | 61300211121              | Würth Elektronik      | CONN HEADER VERT 2POS 2.54MM                                                      |
| 36   | J4,J5,J6,J7,J9,J13,J14,J16,J17,J18,J19,J20,J21,J23,J24,J25,J26,J27,J28,J29,J30,J31 | 22  | Header           | Header_2X1_100MIL       | Regular 100Mil Header |                          |                       |                                                                                   |
| 37   | J8                                                                                 | 1   | CON3             | HDR1X3                  | Default : Pin 1&2     | -                        | -                     | Regular 100Mil Header                                                             |

| Item | Reference                  | Qty | Value            | PCB Footprint                  | Comments                    | Manufacturer Part Number | Manufacturer                             | Description                         |
|------|----------------------------|-----|------------------|--------------------------------|-----------------------------|--------------------------|------------------------------------------|-------------------------------------|
| 38   | J10,J11                    | 2   | PMOD 2x6         | skt_sullins_PPPC062LJB<br>N-RC | -                           | PPPC062LJB-N-RC          | Sullins                                  | CONN HDR 12POS 0.1 GOLD<br>PCB R/A  |
| 39   | J12                        | 1   | Header_2         | Header_2x7                     | Regular<br>100Mil<br>Header | -                        | -                                        | Regular 100Mil Header               |
| 40   | J22                        | 1   | USB TYPE-B RECPT | usb_type_b                     | -                           | GSB3211311WEU            | Amphenol ICC<br>(Commercial<br>Products) | CONN RCPT USB3.0 TYPEB<br>9POS R/A  |
| 41   | J32,J33,J34,J35            | 4   | J-Turret         | TUR_TH                         | -                           | 1573-2                   | Keystone<br>Electronics                  | TERM TURRET SINGLE<br>L=4.72MM TIN  |
| 42   | L1,L2                      | 2   | 220ohm 500mA     | FB0402                         | -                           | MPZ1005S221ET000         | TDK<br>Corporation                       | FERRITE BEAD 220 OHM<br>0402 1LN    |
| 43   | L3                         | 1   | 3.3uH            | 806                            | -                           | SRP2010DPA-3R3M          | Bourns Inc.                              | IND,2X1.6X1MM,3.3UH20%,<br>1.4A,SHD |
| 44   | L6                         | 1   | 1.5uH            | 806                            | -                           | SRP2010DPA-1R5M          | Bourns Inc.                              | IND,2X1.6X1MM,1.5UH20%,<br>2.1A,SHD |
| 45   | PROGRAMN,PB                | 2   | TestPoint_SMT    | TPC32                          | DNL                         | -                        |                                          |                                     |
| 46   | Q1                         | 1   | 2N2222/SOT23     | MMBT2222ALT-1                  | -                           | MMBT2222ALT1HTSA1        | Infineon<br>Technologies                 | TRANS NPN 40V 0.6A SOT-23           |
| 47   | Q2,Q3,Q4,Q5                | 4   | BC817-40-TP      | SOT23-3                        | -                           | BC817-40-TP              | Micro<br>Commercial Co                   | TRANS NPN 45V 0.8A SOT-23           |
| 48   | R1,R2,R3,R35,R37,R38       | 6   | 4.7k             | R0402                          | -                           | RC0402FR-074K7L          | Yageo                                    | RES SMD 4.7K OHM 1%<br>1/16W 0402   |
| 49   | R4,R16,R17,R18,R24,R25,R28 | 7   | 10k              | R0402                          | -                           | RC0402FR-0710KL          | Yageo                                    | RES SMD 10K OHM 1%<br>1/16W 0402    |
| 50   | R5,R13,R19                 | 3   | 2.2k             | R0402                          | -                           | RC0402FR-072K2L          | Yageo                                    | RES SMD 2.2K OHM 1%<br>1/16W 0402   |
| 51   | R6                         | 1   | 22               | R0402                          | -                           | RC0402FR-0722RL          | Yageo                                    | RES SMD 22 OHM 1% 1/16W<br>0402     |
| 52   | R7,R9,R10                  | 3   | 0                | R0402                          | -                           | RC0402JR-070RL           | Yageo                                    | RES SMD 0 OHM JUMPER<br>1/16W 0402  |
| 53   | R14                        | 1   | 12k              | R0402                          | -                           | RC0402FR-0712KL          | Yageo                                    | RES SMD 12K OHM 1%<br>1/16W 0402    |
| 54   | R15                        | 1   | 22               | R0402                          | -                           | RC0402FR-0722RL          | Yageo                                    | RES SMD 22 OHM 1% 1/16W<br>0402     |
| 55   | R20,R29                    | 2   | 4.7K             | R0603                          | -                           | CRCW06034K70FKEA         | Vishay                                   | RES 4.70K OHM 1/10W 1%<br>0603 SMD  |

| Item | Reference                                   | Qty | Value  | PCB Footprint | Comments | Manufacturer Part Number | Manufacturer                    | Description                      |
|------|---------------------------------------------|-----|--------|---------------|----------|--------------------------|---------------------------------|----------------------------------|
| 56   | R21,R22,R45,R64,R67,R75,R82                 | 7   | 330R   | res0603       | -        | ERJ-3EKF3300V            | Panasonic Electronic Components | RES SMD 330 OHM 1% 1/10W 0603    |
| 57   | R23,R30                                     | 2   | 100    | R0402         | -        | ERJ-2RKF1000X            | Panasonic                       | RES SMD 100 OHM 1% 1/10W 0402    |
| 58   | R26,R27,R43,R44,R53,R54,R55,R56,R57,R74,R76 | 11  | 0.1    | R0402         | -        | RL0402FR-070R1L          | Yageo                           | RES 0.1 OHM 1% 1/16W 0402        |
| 59   | R31,R39,R40,R41,R42                         | 5   | 0      | R0402         | -        | RC0402JR-070RL           | Yageo                           | RES SMD 0 OHM JUMPER 1/16W 0402  |
| 60   | R32                                         | 1   | 1k     | R0402         | -        | RC0402FR-071KL           | Yageo                           | RES SMD 1K OHM 1% 1/16W 0402     |
| 61   | R33                                         | 1   | 10K    | R0603         | -        | CRCW060310K0FKEA         | Vishay Dale                     | RES SMD 10K OHM 1% 1/10W 0603    |
| 62   | R34                                         | 1   | 10K    | R0402         | -        | CRCW040210K0JNED         | Vishay Dale                     | RES SMD 10K OHM 5% 1/16W 0402    |
| 63   | R36                                         | 1   | 1k     | R0402         | DNL      | RC0402FR-071KL           | Yageo                           | RES SMD 1K OHM 1% 1/16W 0402     |
| 64   | R46                                         | 1   | DNI    | R0402         | DNL      | ERJ-2GE0R00X             | Panasonic Electronic Components | RES SMD 0 OHM JUMPER 1/10W 0402  |
| 65   | R47,R71,R79,R85                             | 4   | 10K    | res0603       | -        | RC1608F103CS             | Samsung Electro-Mechanics       | RES SMD 10K OHM 1% 1/10W 0603    |
| 66   | R48,R49,R50,R51                             | 4   | 4.7k   | R0402         | DNL      | RC0402FR-074K7L          | Yageo                           | RES SMD 4.7K OHM 1% 1/16W 0402   |
| 67   | R52                                         | 1   | 4.7K   | 402           | -        | RC0402JR-074K7L          | YAGEO                           | RES 4.7K OHM 5% 1/16W 0402       |
| 68   | R58,R90                                     | 2   | 1K     | R0402         | -        | CRCW04021K00JNED         | Vishay                          | RES SMD 1K OHM 5% 1/16W 0402     |
| 69   | R59                                         | 1   | 300R   | 402           | -        | RC0402FR-07300RL         | Yageo                           | RES 300 OHM 1% 1/16W 0402        |
| 70   | R60                                         | 1   | 100R   | 201           | -        | RC0201FR-07100RL         | Yageo                           | RES 100 OHM 1% 1/20W 0201        |
| 71   | R61,R89                                     | 2   | 4.7K   | R0402         | -        | CRCW04024K70JNED         | Vishay                          | RES SMD 4.7K OHM 5% 1/16W 0402   |
| 72   | R62,R83                                     | 2   | 18.2 k | res0402       | -        | RT0402BRD0718K2L         | YAGEO                           | RES SMD 18.2KOHM 0.1% 1/16W 0402 |

| Item | Reference | Qty | Value              | PCB Footprint             | Comments                                      | Manufacturer Part Number | Manufacturer                          | Description                      |
|------|-----------|-----|--------------------|---------------------------|-----------------------------------------------|--------------------------|---------------------------------------|----------------------------------|
| 73   | R63       | 1   | 31.6K              | 603                       | -                                             | RC0603FR-0731K6L         | Yageo                                 | RES 31.6K OHM 1% 1/10W 0603      |
| 74   | R65,R86   | 2   | 12.7 k             | res0402                   | -                                             | ERJ-2RKF1272X            | Panasonic Electronic Components       | RES SMD 12.7K OHM 1% 1/10W 0402  |
| 75   | R66,R87   | 2   | 10K                | 603                       | -                                             | RC0603FR-0710KL          | Yageo                                 | RES SMD 10K OHM 1% 1/10W 0603    |
| 76   | R68       | 1   | 0.01               | R0402                     | -                                             | PE0402FRF070R01L         | Yageo                                 | RES 0.01 OHM 1% 1/16W 0402       |
| 77   | R84       | 1   | 2.61K              | 603                       | -                                             | RC0603FR-072K61L         | Yageo                                 | RES 2.61K OHM 1% 1/10W 0603      |
| 78   | SW1,SW2   | 2   | 430182043816       | 430182043816              | -                                             | 430182043816             | Würth Elektronik                      | SWITCH TACTILE SPST-NO 0.05A 12V |
| 79   | U1        | 1   | ESDR0502N          | ESDR0502N                 | -                                             | ESDR0502NMUTBG           | ON Semiconductor                      | TVS DIODE 5.5VWM 6UDFN           |
| 80   | U2        | 1   | FT2232HL           | tqfp64_0p5_12p2x12p2_h1p6 | -                                             | FT2232HL                 | FTDI                                  | IC USB HS DUAL UART/FIFO 64-LQFP |
| 81   | U3        | 1   | 93LC56C-I/SN       | so8_50_244                | -                                             | 93LC56C-I/SN             | Microchip                             | IC EEPROM 2KBIT SPI 3MHZ 8SOIC   |
| 82   | U4        | 1   | LIFCL-33U-FCCSP104 | FCCSP104                  | Customer Supplied                             | -                        | -                                     | -                                |
| 83   | U5        | 1   | GD25Q128ESIG       | SOP8                      | Customer Supplied                             | GD25Q128ESIGR            | GigaDevice Semiconductor (HK) Limited | IC FLASH 128MBIT SPI/QUAD 8SOP   |
| 84   | U6        | 1   | PI4ULS3V204ZBEX    | TQFN-14                   | -                                             | PI4ULS3V204ZBEX          | Diodes Inc.                           | IC TRANSLTR BIDIRECTIONAL 14TQFN |
| 85   | U7        | 1   | APS25616N-OB9-BG   | BGA                       | Customer Supplied / See Assembly Instructions | APS25616N-OB9-BG         | APMEMORY                              | 256Mb x16 Octal SPI              |
| 86   | U8,U9     | 2   | SP3010-04UTG       | UDFN10_SP3010-04UTG       | -                                             | SP3010-04UTG             | Littelfuse Inc.                       | TVS DIODE 6V 12.3V 10UDFN        |
| 87   | U10,U13   | 2   | AP62201WU-7        | TSOT26_AP62201WU-7        | -                                             | AP62201WU-7              | Diodes Incorporated                   | DCDC CONV HV BUCK TSOT26 T&R 3K  |
| 88   | U11       | 1   | NCP177BMX180TCG    | XDFN4                     | -                                             | NCP177BMX180TCG          | onsemi                                | IC REG LINEAR 1.8V 500MA 4XDFN   |
| 89   | U12       | 1   | NCP145CMX120TCG    | XDFN4                     | -                                             | NCP145CMX120TCG          | onsemi                                | IC REG LINEAR 1.2V 500MA 4XDFN   |

| Item | Reference                    | Qty | Value              | PCB Footprint          | Comments                                                           | Manufacturer Part Number | Manufacturer                    | Description                     |
|------|------------------------------|-----|--------------------|------------------------|--------------------------------------------------------------------|--------------------------|---------------------------------|---------------------------------|
| 90   | X1                           | 1   | CSTNE12M0G55Z000R0 | OSC_CSTNE12M0G55Z000R0 | -                                                                  | CSTNE12M0G55Z000R0       | Murata Electronics              | CERAMIC RES 12.0000MHZ 33PF SMD |
| 91   | Y1                           | 1   | LFSPX0076024REEL   | 4-SMD                  | -                                                                  | LFSPX0076024REEL         | IQD Frequency Products          | XTAL OSC XO 25.0000MHZ CMOS SMD |
| 92   | Y2                           | 1   | 50 MHz             | 6SMD                   | -                                                                  | XLL736050.000000I        | Renesas Electronics America Inc | XTAL OSC XO 50.0000MHZ LVDS SMD |
| 93   | 12MHZ                        | 1   | T POINT R          | TP                     | DNL                                                                | -                        | -                               | -                               |
| 94   | Shunt for headers            | 24  | SPC02SYAN          | -                      | Load the shunts mentioned in the BoM & the remaining are Bag & Tag | SPC02SYAN                | Sullins Connector Solutions     | CONN JUMPER SHORTING GOLD FLASH |
| 95   | LIFCL-33U-EVN BOARD REVA PCB | 1   | -                  | -                      | -                                                                  | 305-PD-22-0777           | PACTRON                         | -                               |

**Note:**

- For the initial build of 25 boards, please use the 15 units of APS512XXN-OBX9-BG & 10 units of APS256XXN-OB9-BG @ location U7

## References

For more information, refer to:

- [Lattice Sales Office](#)
- [Lattice Insights](#) for Lattice Semiconductor training courses and learning plans
- [CrossLink-NX-33 and CrossLinkU-NX Data Sheet \(FPGA-DS-02104\)](#)
- [CrossLinkU-NX Web Page](#)
- [sysCONFIG Usage Guide for Nexus Platform \(FPGA-TN-02099\)](#)
- [Lattice Radiant](#) FPGA design software
- [Lattice Boards and Kits](#)

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For frequently asked questions, refer to the Lattice Answer Database at [www.latticesemi.com/en/Support/AnswerDatabase](http://www.latticesemi.com/en/Support/AnswerDatabase).

## Revision History

### Revision 1.0, October 2023

| Section | Change Summary   |
|---------|------------------|
| All     | Initial release. |





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